

1. Fundamentals of Automation Testing.

- Automation Testing is the process of using software tools (like selenium) to run tests on applications automatically.
- It checks whether the actual output matches the expected output, without manual efforts each time.
- Example: Instead of clicking buttons or entering data manually every time, a script does it for you.
- **Benefits of Automation Testing:**
 - a) Saves time
 - b) Reduces human errors
 - c) Fast execution
 - d) Good for repeated tests

2. Introduction of Functional Testing Using Automation Tool.

- Functional testing checks if the application features work correctly.
- Automation tools are used to test functions automatically.
- Scripts perform actions like login, search, and button clicks.
- Helps in fast, accurate, and repeated testing.
- Common tools: Selenium, Appium, TestNG.

3. Selenium IDE with use of Fire bug tools.

Selenium IDE

- Selenium IDE is a **browser extension** used for recording and playing back test cases.
- It is mainly used for **quick automation** and **beginner-level testing**.
- Test steps are recorded automatically as you click on the web page.
- You can **edit, save, and run** recorded scripts easily.

Firebug Tool

- Firebug was a **Firefox add-on** used for inspecting web elements.
- It helped testers find **ID, Name, Class, XPath** of UI elements.
- Using Firebug, you could **locate elements** quickly for writing Selenium scripts.
- Firebug is no longer active, but its features moved into **Firefox Developer Tools**.

How Selenium IDE and Firebug Worked Together

- Use Firebug to **inspect an element** and get its locator.
- Use that locator inside Selenium IDE for **recorded or manual steps**.
- This helped testers create **accurate and reliable** automation test cases.

4. Introduction of Non-Functional Testing Using Automation Tool.

- Non-functional testing checks the quality of an application, not the features.
- It focuses on performance, speed, load, security, and usability.
- Automation tools help to test how the system behaves under stress or heavy load.
- Tools like JMeter, LoadRunner, and Postman are used for non-functional automation.
- Automated non-functional tests give fast, accurate, and repeatable results.
- It ensures the application is stable, efficient, and reliable for real users.

5. Introduction about Load Runner-up.

- LoadRunner is a performance testing tool used to check how an application behaves under load.
- It helps testers measure speed, stability, and scalability of a system.
- The tool creates multiple virtual users to test how the application performs with many users at the same time.
- LoadRunner identifies bottlenecks, slow response time, and performance issues.
- It supports testing for web, mobile, and enterprise applications.
- LoadRunner provides detailed performance reports for analysis and improvement.

6. Load Runner-up procedure.

1. Requirement Analysis

Understand what to test: number of users, scenarios, and performance goals.

2. Create Vuser Script (Using VuGen)

Record user actions on the application.

Add checkpoints, transactions, and validations.

Save the script for load execution.

3. Design Scenario (Using Controller)

Set number of virtual users.

Set ramp-up time, test duration, and load pattern.

Add test scripts to the scenario.

4. Execute the Load Test

Start the test and monitor server/system performance.

Observe response time, throughput, errors, CPU, and memory usage.

5. Analyze Results (Using Analysis Tool)

Review graphs and reports.

Identify bottlenecks and performance issues.

Share findings with the development team.

7. How to use Load Runner-up.

LoadRunner is a performance testing tool used to check how an application behaves under different loads — for example, how your website or app performs when 1000 users access it at once.

How to use load Runner up

1. Install LoadRunner

- Install LoadRunner and open VuGen, Controller, and Analysis components.

2. Create a Script in VuGen

- Open VuGen (Virtual User Generator).
- Select a protocol (like Web HTTP/HTML).
- Record the steps you perform on the application.
- Enhance the script with checkpoints, loops, and validations.
- Save the script.

3. Design Load Scenario in Controller

- Open Controller.
- Add the Vuser script you created.
- Set number of virtual users.
- Set load pattern (Ramp-up, Steady load, Ramp-down).
- Configure test duration and execution settings.

4. Run the Load Test

- Click Start to execute the scenario.

- Monitor performance metrics like response time, throughput, CPU, and memory.

5. Analyze Test Results

- After execution, open Analysis Tool.
- View graphs, charts, and reports.
- Identify performance issues and bottlenecks.

8. Load Runner-up IDE.

- LoadRunner IDE mainly refers to VuGen (Virtual User Generator), which is the scripting environment of LoadRunner.
- VuGen IDE is used to record, edit, and enhance performance test scripts.
- It provides features like recording panel, code editor, runtime settings, and logs.
- The IDE helps testers add transactions, checkpoints, loops, and correlations.
- It supports multiple protocols like Web HTTP, Web Services, SAP, Oracle, etc.
- VuGen IDE also allows debugging and replay of scripts before running load tests.

9. Generate Single Scripting and Assign V-Users Multi Scripting with Grouping.

1) Generate Single Scripting

- Create only one script in LoadRunner using VuGen.
- Record all required steps (login → actions → logout).
- Enhance the script with transactions, checkpoints, and correlations.
- Save and test the script by replaying it.
- This single script will be used for all virtual users.

2) Assign V-Users (Virtual Users)

- Open Controller in LoadRunner.
- Add the single script you created.
- Set the number of V-users (example: 10, 50, 100).
- Configure ramp-up time and duration.
- Assign how many V-users will run the script at the same time.

3) Multi Scripting

- Create multiple scripts for different scenarios.

Example:

- Script 1 → Login
 - Script 2 → Search
 - Script 3 → Add to Cart
- Add all scripts in the Controller.

- Each script can have different configurations and number of V-users.

4) Grouping

- In Controller, create Groups to organize scripts.
Example:
 - Group A → 20 V-users (Login Script)
 - Group B → 30 V-users (Search Script)
 - Group C → 10 V-users (Cart Script)
- Grouping helps run different user behaviors together.
- It gives realistic load and easy management in reports.

10. Parameterization in Load Runner-up.

- Parameterization in LoadRunner is the process of making a script more flexible and reusable by replacing hard-coded values with variables.
- This allows you to simulate a wider range of user behavior, and to run the same script with different sets of data.

11. Multi Scripting with Grouping.

"Multi Scripting with Grouping" refers to the practice of organizing and executing multiple scripts, often in a structured or hierarchical manner, to achieve a larger objective. This approach offers benefits such as modularity, reusability, and easier management of complex tasks.